

Svarun Soda

Los Angeles, CA - USA

☎ (818) 852 3202 • ✉ runsva2001@gmail.com • in svarunsoda • SvarunSoda

Professional Summary

Computer Science M.S. graduate with research experience in unconventional computational models & analog number-processing paradigms. Hands-on experience at NASA JPL, CalTech, and Medtronic developing software tools and conducting experimental testing, focused in AI-based image recognition, photonic circuit programming, software-in-the-loop simulation and medical firmware verification.

Education

California State University, Northridge

M.S. – Computer Science

Northridge, CA

Aug 2024 – May 2026

○ GPA: 3.86 / 4.0

○ Thesis: *Phase-based Oscillatory Computation (PHOSCO)*

○ Developed a novel oscillator-based temporal computational framework for continuous value-space arithmetic operations.

California State University, Northridge

B.S. – Computer Science

Northridge, CA

Aug 2022 – May 2024

○ GPA: 3.65 / 4.0

○ Award: Pearl Schuman Bloch Memorial Scholarship (2023)

College of the Canyons

A.S. – Computer Science

Santa Clarita, CA

Aug 2020 – Jun 2022

○ GPA: 3.34 / 4.0

Experience

Medtronic Minimed

Software Verification Co-Op

Northridge, CA

Sep 2024 – May 2026

○ Developed and executed automated testing frameworks for insulin delivery firmware using C# and JavaScript.

○ Integrated software tools with embedded hardware to validate reliability and compliance.

○ Identified and isolated firmware issues through accelerated verification testing.

NASA Jet Propulsion Laboratory

AEGIS Machine Learning Intern

Pasadena, CA

Jun 2024 – Aug 2024

○ Enhanced *Perseverance* Mars rover onboard image-recognition performance by 29% through ML model training.

○ Built modular Python pipelines using NASA's vision tools and open-source APIs.

○ Supported development of ground-truth labeling workflows for autonomous imaging systems.

California Institute of Technology

Summer Research Intern – Exoplanet Laboratory

Pasadena, CA

Jun 2023 – Aug 2023

○ Programmed and interfaced with optical devices for the HISPEC spectrograph project (Keck II).

○ Built Python-based serial and TCP tools for data acquisition and light throughput analysis.

○ Collaborated with researchers on testing optical fiber switching hardware.

California Institute of Technology

Software Developer Intern – Exoplanet Laboratory

Pasadena, CA

Sep 2022 – Jan 2023

○ Developed flattening algorithm for *Dynamic Spectrum Flattener* photonic chip.

○ Processed and visualized light spectra using Python, NumPy, and Matplotlib.

○ Co-authored astrophotonics research papers published at SPIE conferences.

Projects

California State University, Northridge

Senior Design Project – EDAmame

Northridge, CA

Aug 2023 – Jun 2024

○ Contributed to an open-source Electronic Design Automation (EDA) software suite.

○ Implemented Java modules using IntelliJ and GitHub version control.

College of the Canyons

NASA RockSat-X and HASP Projects

Santa Clarita, CA

Nov 2021 – Sep 2024

○ Designed and implemented embedded control and data acquisition systems for NASA payload missions.

○ Assisted with payload integration and testing during RockSat-X launch operations at Wallops Flight Facility.

Technical Skills

Programming: C++, C, Python, JavaScript, C#, Java, Lua

Tools: Visual Studio, Arduino IDE, KiCad, AutoCAD, IntelliJ

Specialties: Embedded Systems, Machine Learning, Analog Computing, Data Visualization